Installing Oracle Database 19c on Oracle Linux

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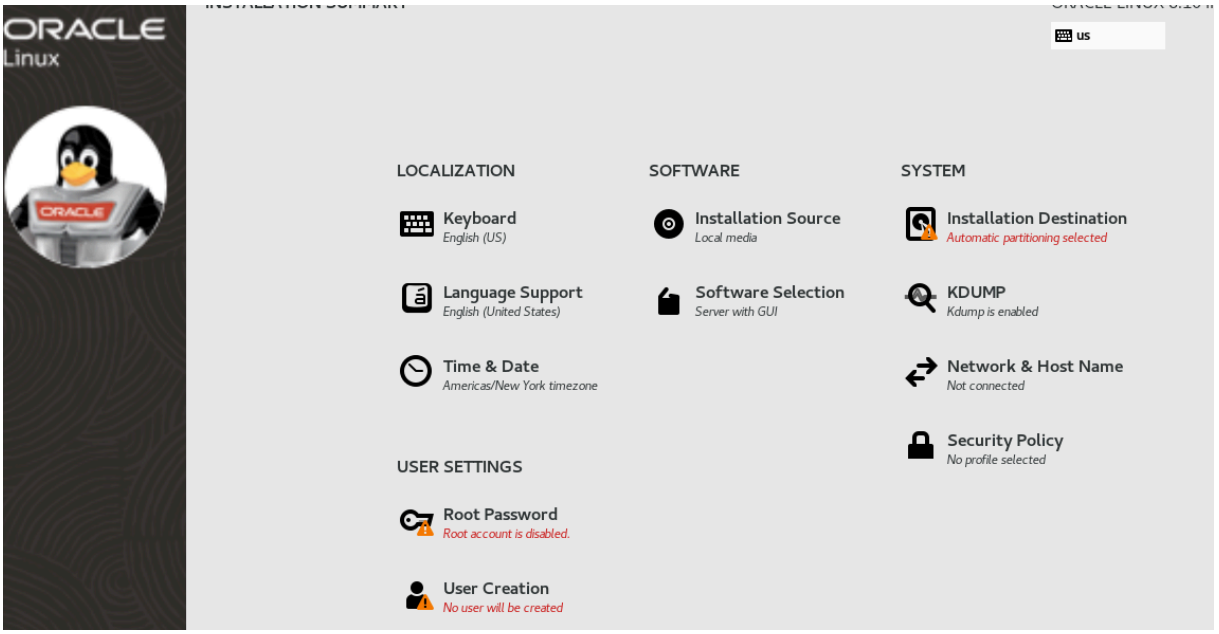
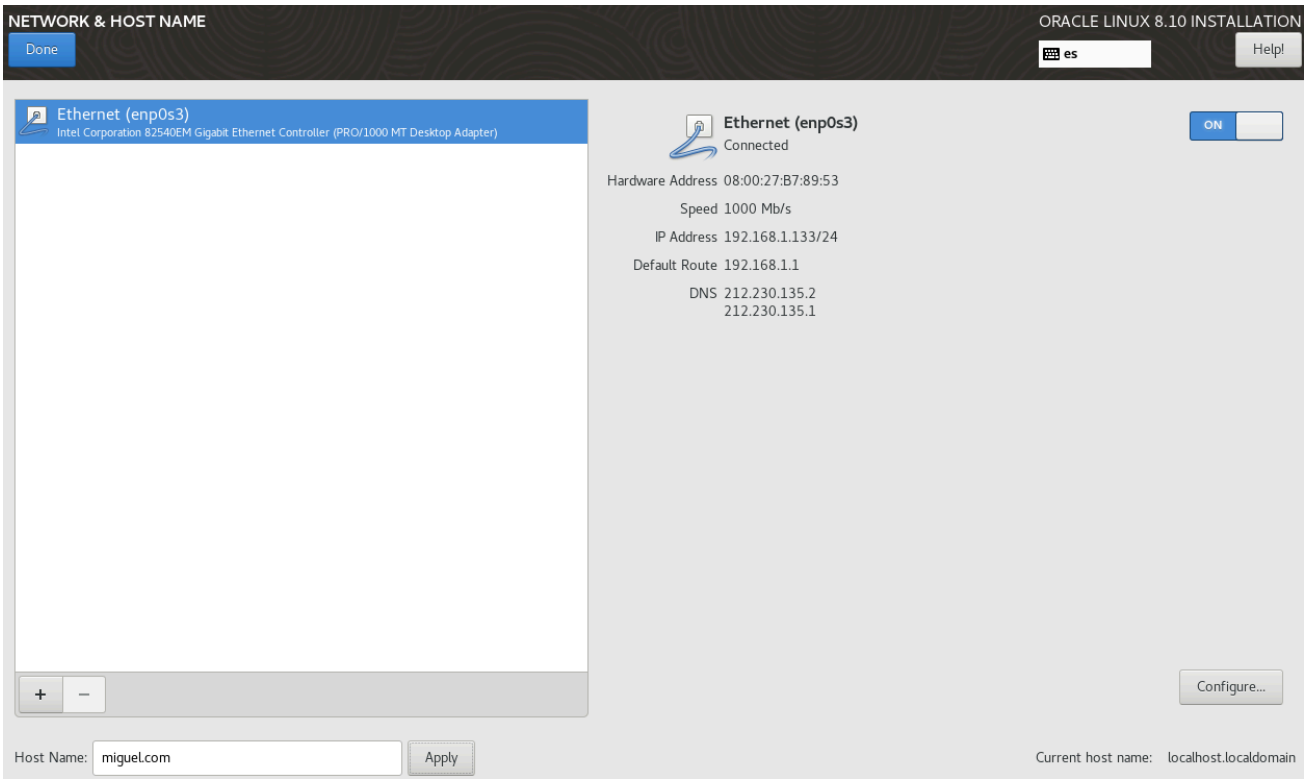
1.Oracle Linux

First, we will download an Oracle Linux distribution to install Oracle Database 19c on it. We'll download it from the website you can find in the bibliography section.

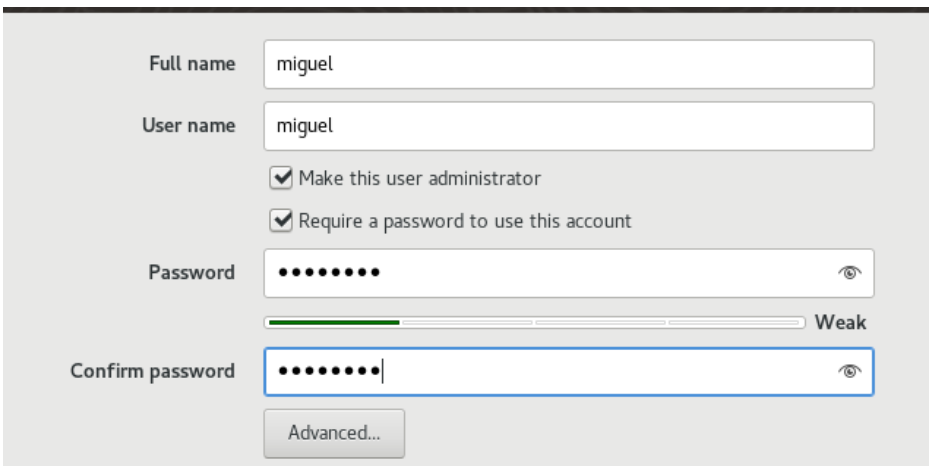


Let's configure Oracle Linux. We are going to configure the Time and Date, Keyboard Distribution, Software Selection(we will choose Server with GUI and Development Tools) and we will disable Security Policy to avoid problems during the installation.

Then we enable the Ethernet and change the host name, this one will be important later.



To finish we will set up a password for the root user and create a new user.



Once everything is configured we will install the guest additions. We will be able to copy and paste between the two machines, see the virtual box in full-screen and so on.

2.Preparing Linux for the database installation

There are some prerequisites or steps we have to accomplish in order to install Oracle DB.

1. Edit the host file by using vi /etc/hosts and add:
   <IP-address> <hostname.domain> <hostname>

```
[miguel@miguel ~]$ su root
Password:
[root@miguel miguel]# hostname
miguel.com
[root@miguel miguel]# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.133  netmask 255.255.255.0  broadcast 192.168.1.255
```

```
miguel@miguel:/home/miguel
File  Edit  View  Search  Terminal  Help
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.1.133   miguel.com      miguel
```

2. Automatic preinstall. This package performs a lot of tasks such as create the Oracle user who will be vital to install Oracle Database.

```
[root@miguel miguel]# dnf install -y oracle-database-preinstall-19c
```

```
[root@miguel miguel]# id oracle
uid=54321(oracle) gid=54321(oinstall) groups=54321(oinstall),54322(dba),54323(oper)
,54324(backupdba),54325(dgdba),54326(kmdba),54330(racdba)
```

3. Create directories and grant permissions to Oracle user.

```
[root@miguel miguel]# mkdir -p /u01/app/oracle/product/19c/dbhome_1
```

u01 app oracle product 19c dbhome_1

```
[root@miguel miguel]# mkdir -p /u01/app/oracle/product/19c/dbhome_1
[root@miguel miguel]# mkdir -p /u02/oradata
[root@miguel miguel]# chown -R oracle:oinstall /u01 /u02
[root@miguel miguel]# chmod -R 775 /u01 /u02
```

4. Only because we are in a student/learning environment we will disable the firewall. In real life we won't ever do this.

```
[root@miguel miguel]# systemctl stop firewalld
[root@miguel miguel]# systemctl disable firewalld
Removed /etc/systemd/system/multi-user.target.wants/firewalld.service.
Removed /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service.
```

5. Changing the password for the user oracle that was created with the preinstall command.

```
[root@miguel miguel]# passwd oracle
```

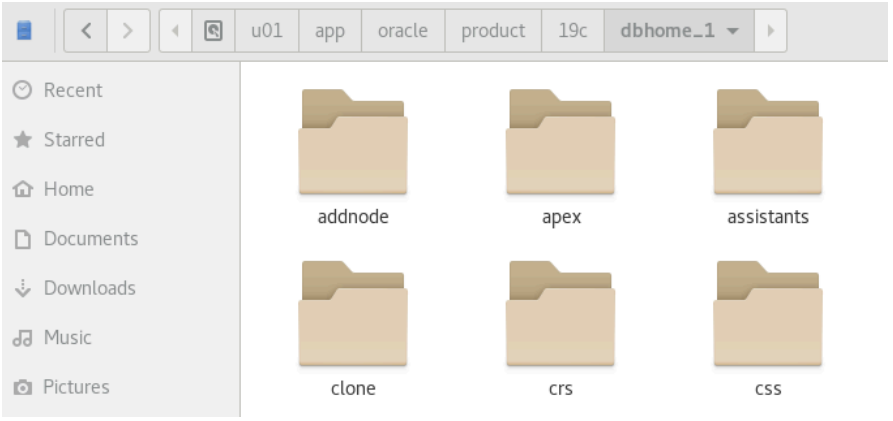
6. Creating a file/script with environment variables crucial for Oracle Database and adding it to the Oracle bash_profile to be executed automatically each time the user Oracle log in.

```
[root@miguel miguel]# cat > /home/oracle/scripts/setEnv.sh <<EOF
> export TMP=/tmp
> export TMPDIR=\$TMP
>
> export ORACLE_HOSTNAME=miguel.com
> export ORACLE_UNQNAME=cdb1
> export ORACLE_BASE=/u01/app/oracle
> export ORACLE_HOME=\$ORACLE_BASE/product/19c/dbhome_1
> export ORA_INVENTORY=/u01/app/oraInventory
> export ORACLE_SID=cdb1
> export PDB_NAME=pdb1
> export DATA_DIR=/u02/oradata
>
> export PATH=/usr/sbin:/usr/local/bin:\$PATH
> export PATH=\$ORACLE_HOME/bin:\$PATH
>
> export LD_LIBRARY_PATH=\$ORACLE_HOME/lib:/lib:/usr/lib
> export CLASSPATH=\$ORACLE_HOME/jlib:\$ORACLE_HOME/rdbms/jlib
> EOF
```

```
[root@miguel miguel]# echo ". /home/oracle/scripts/setEnv.sh" >> /home/oracle/.bash_profile
```

3.Installing Oracle Database 19c

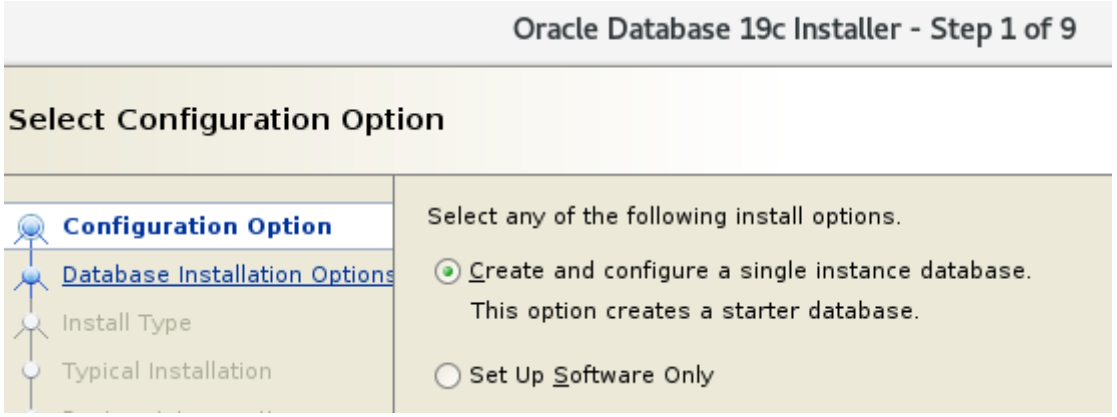
Now, we are going to install Oracle Database, to carry out this task we will login as an Oracle user. We will download the zip archive from the official Oracle website. Then, we will unzip the archive in the Oracle Home directory.

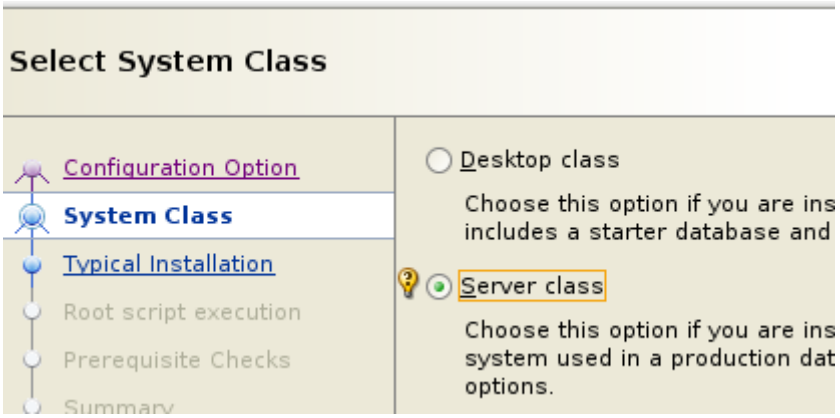


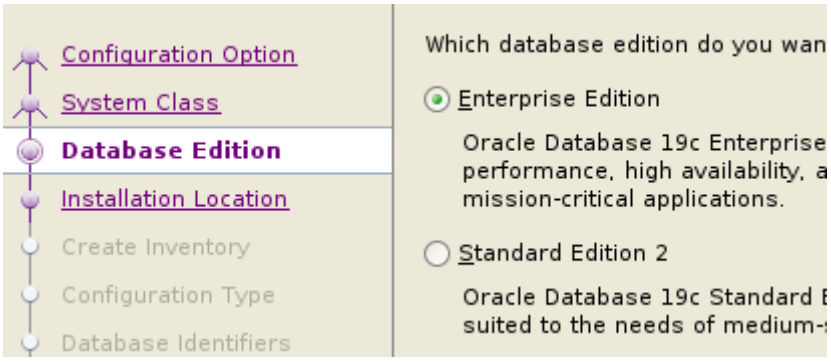
Then we will execute the following commands:

```
[oracle@miguel dbhome_1]$ export CV_ASSUME_DISTID=OEL7.6
[oracle@miguel dbhome_1]$ ./runInstaller
Launching Oracle Database Setup Wizard...
```

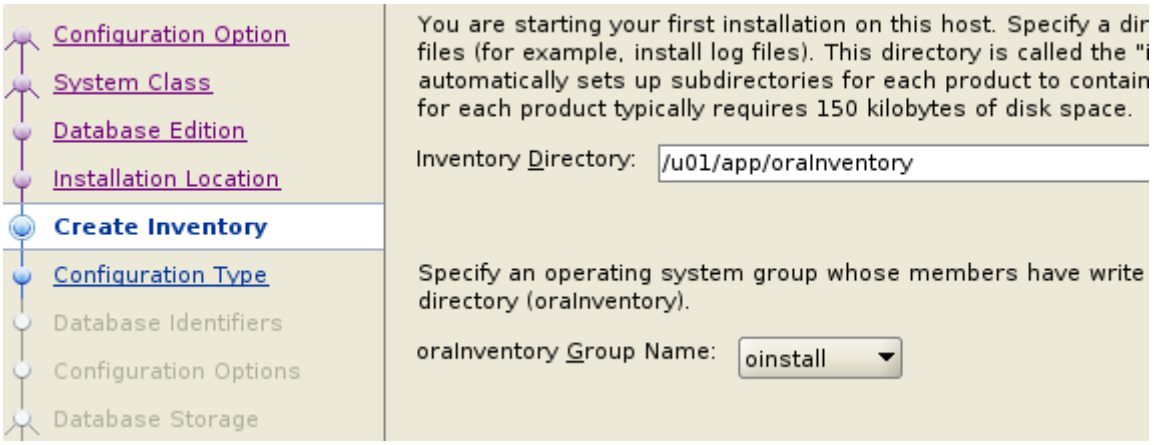
The Oracle Database 19c Installer will open and we will go through the following windows/steps:




Select System Class
Configuration Option
System Class
Typical Installation
Root script execution
Prerequisite Checks
Summary
Desktop class
Choose this option if you are ins
includes a starter database and
Server class
Choose this option if you are ins
system used in a production dat
options.


Configuration Option
System Class
Database Edition
Installation Location
Create Inventory
Configuration Type
Database Identifiers
Which database edition do you wan
Enterprise Edition
Oracle Database 19c Enterprise
performance, high availability, a
mission-critical applications.
Standard Edition 2
Oracle Database 19c Standard E
suited to the needs of medium-

Configuration Option
System Class
Database Edition
Installation Location
Create Inventory
Configuration Type
Specify a path to place all Oracle software and configuration-relat
installation owner. This location is the Oracle base directory for th
Oracle base:
/u01/app/oracle
This software directory is the Oracle Database home directory.
Software location: /u01/app/oracle/product/19c/dbhome_1


Configuration Option
System Class
Database Edition
Installation Location
Create Inventory
Configuration Type
Database Identifiers
Configuration Options
Database Storage
You are starting your first installation on this host. Specify a dir
files (for example, install log files). This directory is called the "i
automatically sets up subdirectories for each product to contain
for each product typically requires 150 kilobytes of disk space.
Inventory Directory:
/u01/app/oraInventory
Specify an operating system group whose members have write
directory (oraInventory).
oraInventory Group Name:
oinstall

Configuration Option
System Class
Database Edition
Installation Location
Create Inventory
Configuration Type
Database Identifiers

Select the type of database that you want to create.

General Purpose / Transaction Processing

A starter database designed for general purpose use, or for trans

Data Warehousing

A starter database optimized for data warehousing applications.

Configuration Option
System Class
Database Edition
Installation Location
Create Inventory
Configuration Type
Database Identifiers
Configuration Options
Database Storage

Provide the identifier information required to access the database un
uniquely identified by a Global database name, typically of the form "
referenced by at least one Oracle instance which is uniquely identifie
this computer by an Oracle system identifier (SID).

Global database name: orcl.com

Oracle system identifier (SID): cdb1

Create as Container database

Creates a database container for consolidating multiple database
and enables database virtualization.

Pluggable database name: orclpdb

Configuration Option
System Class
Database Edition
Installation Location
Create Inventory
Configuration Type
Database Identifiers
Configuration Options

Memory | Character sets | Sample schemas

You can choose to install sample schema in the s
option, Human Resources schema will be installe
SYSAUX tablespace.

Install sample schemas in the database

Configuration Option
System Class
Database Edition
Installation Location
Create Inventory
Configuration Type
Database Identifiers
Configuration Options
Database Storage

File system

Use a file system for database storage. For optimal databas
Oracle recommends that you install data files and the Oracle

Specify database file location: /u01/app/oracle/oradata

Oracle Automatic Storage Management

Oracle Automatic Storage Management (Oracle ASM) simplifie
and places database files for optimal I/O performance. Selec
Oracle ASM.

Recovery Options
Schema Passwords
Operating System Groups

Password: •••••• Confirm password: ••••••

Perform Prerequisite Checks

19c ORACLE Database

- Configuration Option
- System Class
- Database Edition
- Installation Location
- Create Inventory
- Configuration Type
- Database Identifiers
- Configuration Options
- Database Storage
- Management Options
- Recovery Options
- **Schema Passwords**
- Operating System Groups
- Root script execution
- **Prerequisite Checks**
- Summary
- Install Product
- Finish

Verification Result

Some of the minimum requirements for installation are not completed. Review and fix the issues listed in the following table, and recheck the system.

Check Again | Fix & Check Again | Show Failed | Ignore All

Checks	Status	Fixable
Checks		
Swap Size	Ignored	No

This is a prerequisite condition to test whether sufficient total swap space is available on the system. (more details)
Expected Value : 6.9269GB (7263408.0KB)
Actual Value : 6.0391GB (6332412.0KB)

Help | < Back | Next > | Install | Cancel

Execute Configuration Scripts

The following script needs to be executed as the "root" user.

Scripts
/u01/app/oraInventory/orainstRoot.sh
/u01/app/oracle/product/19c/dbhome_1/root.sh

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Finish

19c ORACLE Database

- Configuration Option
- System Class
- Database Edition
- Installation Location

The configuration of Oracle Database was successful.

Note:

Oracle Enterprise Manager Database Express URL: https://miguel.com:5500/em

To conclude, we are going to check if the database was installed correctly.

```
[oracle@miguel ~]$ sqlplus

SQL*Plus: Release 19.0.0.0.0 - Production on Fri Mar 13 12:14:03 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.

Enter user-name: sys as sysdba
Enter password:

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

4.Errors and troubles

Before execute the ./runInstaller we have to run: export CV_ASSUME_DISTID=OEL7.6 in order to avoid the following error while installing Oracle Database, [INS-08101] Unexpected error while executing the action at state: 'supportedOSCheck'. The command export CV_ASSUME_DISTID=OEL7.6 tricks the Oracle Database installer and makes it believe the OS is Oracle Linux 7.6 instead of 8.10.

5.Resources

Download Oracle Linux 8.10:
https://yum.oracle.com/oracle-linux-isos.html

How to install Oracle Database 19c on Oracle Linux:
https://oracle-base.com/articles/19c/oracle-db-19c-installation-on-oracle-linux-8

Download Oracle Database 19c:
https://www.oracle.com/es/database/technologies/oracle19c-linux-downloads.html